

Shahriar Alam Patwary

Computer Science & Engineering, Bangladesh University of Engineering and Technology (BUET)

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PROFILE

Third-year CSE undergraduate at BUET interested in **AI/ML, systems, embedded systems, research, and technology-driven product development**. Experienced in building software and hardware prototypes, developing systems used by real organizations, and working on technical projects in small teams. Interested in transforming **research and experimentation into functional prototypes and real-world products**.

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

B.Sc. in Computer Science and Engineering

CGPA: **3.97 / 4.00**

Dhaka, Bangladesh

Expected Graduation: July 2028

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Discrete Mathematics, Calculus, Linear Algebra, Probability & Statistics, Artificial Intelligence, Operating Systems, Compiler, Computer Architecture, Microcontrollers & Embedded Systems.

SELECTED PROJECTS & ACHIEVEMENTS

AgriSense — Agentic AI Platform

Champion, IUT 12th ICT Fest 2026

- Developed an agentic AI platform for agricultural decision support in a **3-member team**; competed among **20 finalists selected from 400+ registrations**.
- Built workflows for crop recommendation, seasonal planning, and financial projections using LLMs, vector search, weather, SMS, voice, and payment services.
- Tech:** React, Node.js, Express, Supabase, LanceDB, OpenAI, Gemini, Groq, OpenRouter.

BADHAN Report Management System

Production Software

- Developed a digital reporting platform currently used by **180+ BADHAN units nationwide**.
- Built reporting workflows, role-based access, audit logging, bilingual interfaces, automated PDF generation, and deployment infrastructure.
- Tech:** React, Node.js, Express, PostgreSQL, Docker, Puppeteer.

Desktop Companion Buddy — CSE 316 Project

Microcontroller & Microprocessor Sessional

- Developing an interactive desktop companion robot using the **ATmega32**, DC motors, servos, IR cliff detection, ultrasonic obstacle detection, and an OLED display.
- Implemented responsive movement and expressive eye animations using timers, PWM, interrupts, sensor interfacing, actuator control, and embedded C.

Dheu — NASA Space Apps Challenge 2025

National Selection

- Developed a project using **SAR satellite data to observe ocean health**; advanced from the local round to **national-level selection**.
- Built a functional prototype as part of a team under strict competition constraints.

LEADERSHIP & ACTIVITIES

BUET Robotics Society

Deputy Head of Media & Outreach Team

Involved in organizing robotics competitions, workshops, technical programs, and student initiatives; contributed to event planning, communications, partnerships, and coordination.

TECHNICAL SKILLS

Languages C, C++, Python, Java, JavaScript, TypeScript, Bash

AI/ML Machine Learning, LLM APIs, Agentic AI, RAG, Vector Search, Computer Vision

Systems Linux, xv6, Operating Systems, ANTLR4, Compiler Construction

Web React, Node.js, Express, PostgreSQL, Supabase, REST APIs

Embedded ATmega32, AVR C, Timers, PWM, ADC, Interrupts, UART, I²C, SPI, Sensors & Actuators

Tools Git, GitHub, Docker, VS Code, WSL/Linux

RESEARCH & INNOVATION INTERESTS

AI/ML • AI Systems • Multimodal & Agentic AI • Systems Research
Embedded Intelligence & Robotics • Quantum Computing • Technology Entrepreneurship